POPULATION HEALTH INFORMATICS PORTFOLIO

Closing diabetes care gaps: from community need to registry-driven outreach

Where in Palm Beach and Glades counties is diabetes need greatest, what stands in the way of care in each place, and how can a registry with AI decision support turn that into outreach?

Open the needs dashboard Open the project plan

Key takeaways

1. **Need follows insurance and income, not only distance.** Of the 15 top-ranked Palm Beach tracts, 6 are western and 9 are urban pockets in Lake Worth Beach, West Palm Beach and Westgate. 13 of the 15 rank high mainly on social vulnerability.

2. **Two places, two different barriers.** Western Palm Beach is held back by cost (27.0% of adults 18-64 uninsured, 46.2% of residents below 150% of the poverty line). Glades is held back by distance (14.7 miles to a health center site, against 3.0 in the east).

3. **Diabetes is more common in the west and in Glades than in the east.** 19.5% of adults in western Palm Beach and 17.5% in Glades County have diagnosed diabetes, against 13.1% in eastern and central Palm Beach.

4. **The ranking holds up, the exact top 10 does not.** Across five alternative weightings, rank correlation stays between 0.92 and 0.99, but only 5 to 10 of the top 10 tracts stay put. Use the top tier, not a single rank.

5 **AI helped modestly and unevenly (synthetic patients).** The risk model beat a simple rule by 0.05 AUC, yet caught 29% of later ED visits among privately insured patients against 56% among uninsured patients, and was no better than chance for Asian and other patients. Audit first, then deploy.

6 **Use real data for place and synthetic data for method.** The synthetic areas look alike on care gaps, so the rural story comes from public data. The screening scenario illustrates a formula and predicts nothing.

Approach

I joined public data (CDC PLACES 2025, CDC social vulnerability index 2022, USDA rural-urban codes, HRSA health center sites) into a tract-level needs dashboard that compares three areas: eastern and central Palm Beach, western Palm Beach and Glades County. Western Palm Beach is the populated tracts in the federal farmworker shortage designation around Belle Glade and Pahokee.

I then built a diabetes registry prototype on synthetic patients, with four HEDIS-style care-gap measures, a risk model, an equity audit and a retinopathy-screening scenario.

Part 1: the community needs PUBLIC DATA

Three areas compared

MEASURE	EASTERN AND CENTRAL PALM BEACH	WESTERN PALM BEACH	GLADES COUNTY
Residents	1,458,783	33,392	12,126
Adults with diagnosed diabetes	13.1%	**19.5%**	17.5%
Uninsured, ages 18-64	14.2%	**27.0%**	23.4%
Residents below 150% of poverty line	18.7%	**46.2%**	36.2%
Households with no vehicle	5.5%	**18.6%**	2.0%
Average miles to nearest health center site	3.0	3.6	**14.7**
Residents more than 5 miles from a site	16%	29%	**100%**
Permanent health center sites	24	2	1

Population-weighted averages of tract values. PLACES values are model-based estimates. Distance is a straight line from the tract's interior point to the nearest permanent HRSA site. County-level, Glades County's age-adjusted diabetes rate is 14.4% (95% interval 12.5 to 16.4) against 10.6% (9.3 to 12.1) for Palm Beach County.

Top 10 Palm Beach tracts for outreach

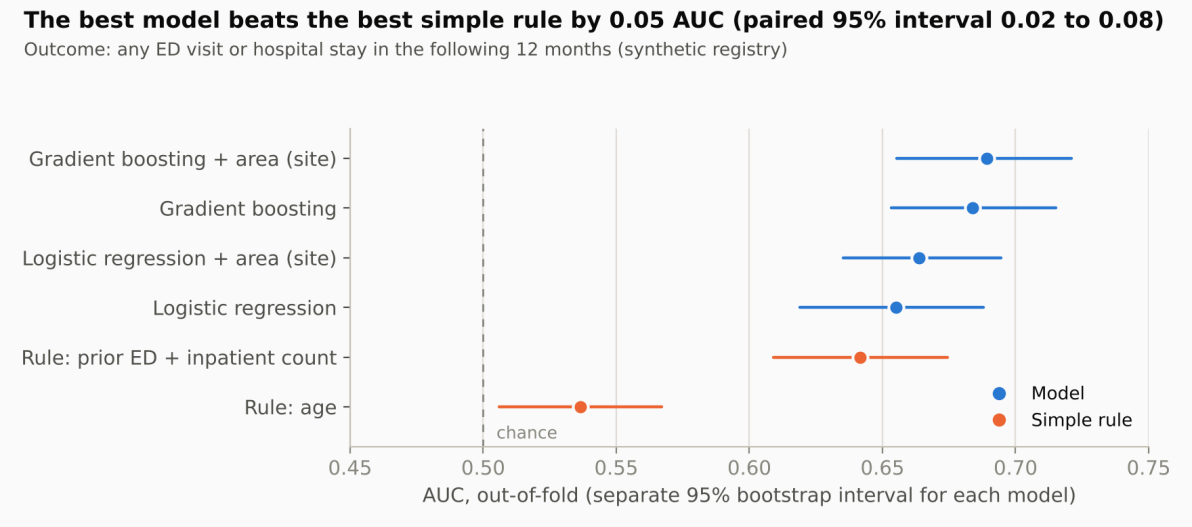
The priority score averages three percentile ranks, one third each: diabetes burden, social vulnerability, and access gap (uninsured adults, no routine checkup, households without a vehicle). Only Palm Beach tracts are scored.

RANK	COMMUNITY AND TRACT	DIABETES	UNINSURED 18-64	NO VEHICLE	MILES TO SITE	SCORE	MAIN DRIVER
1	Unincorporated western Palm Beach, Tract 80.01	23.8%	35.2%	32.9%	11.3	97.7	Social vulnerability
2	Lake Worth Beach, Tract 44.02	16.8%	38.6%	23.1%	2.0	94.4	Social vulnerability
3	West Palm Beach, Tract 16	19.3%	28.9%	16.1%	1.0	94.2	Social vulnerability
4	Westgate, Tract 29	16.6%	41.5%	17.4%	0.1	93.5	Social vulnerability
5	South Bay, Tract 83.02	17.5%	27.8%	13.5%	2.7	92.6	Social vulnerability
6	Unincorporated western Palm Beach, Tract 82.03	17.4%	28.4%	14.7%	2.2	92.4	Social vulnerability
7	West Palm Beach, Tract 22	23.3%	24.8%	31.4%	0.5	91.8	Social vulnerability
8	Lake Worth Beach, Tract 51.02	16.7%	32.8%	10.6%	1.8	91.3	Social vulnerability
9	Belle Glade, Tract 82.02	26.9%	25.3%	18.8%	0.8	90.7	Diabetes burden
10	Schall Circle, Tract 19.09	16.1%	30.5%	11.3%	1.3	90.3	Social vulnerability

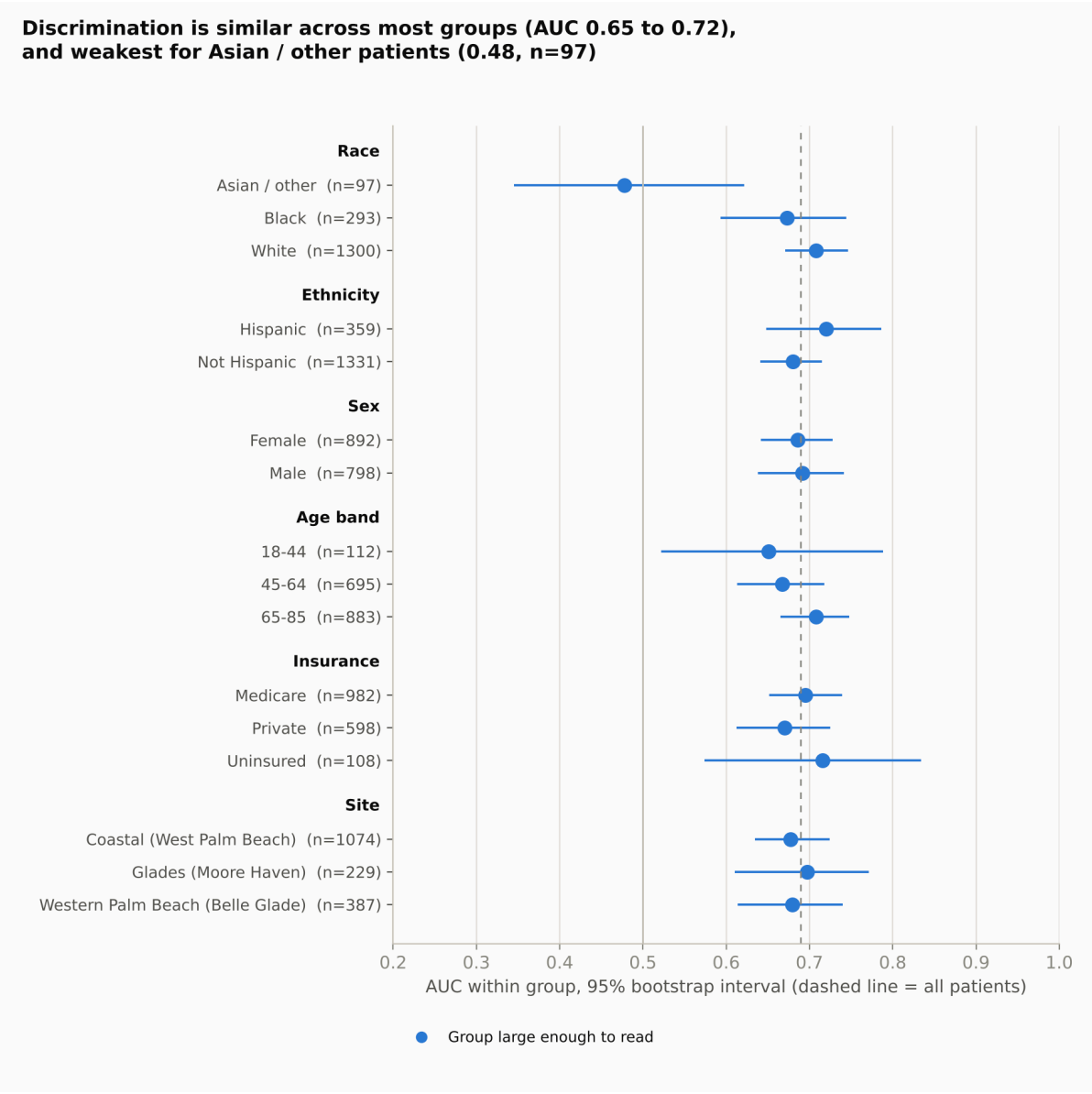
The dashboard lists the top 15 and shows every tract on a map.

Part 2: the registry and the AI SYNTHETIC PATIENTS

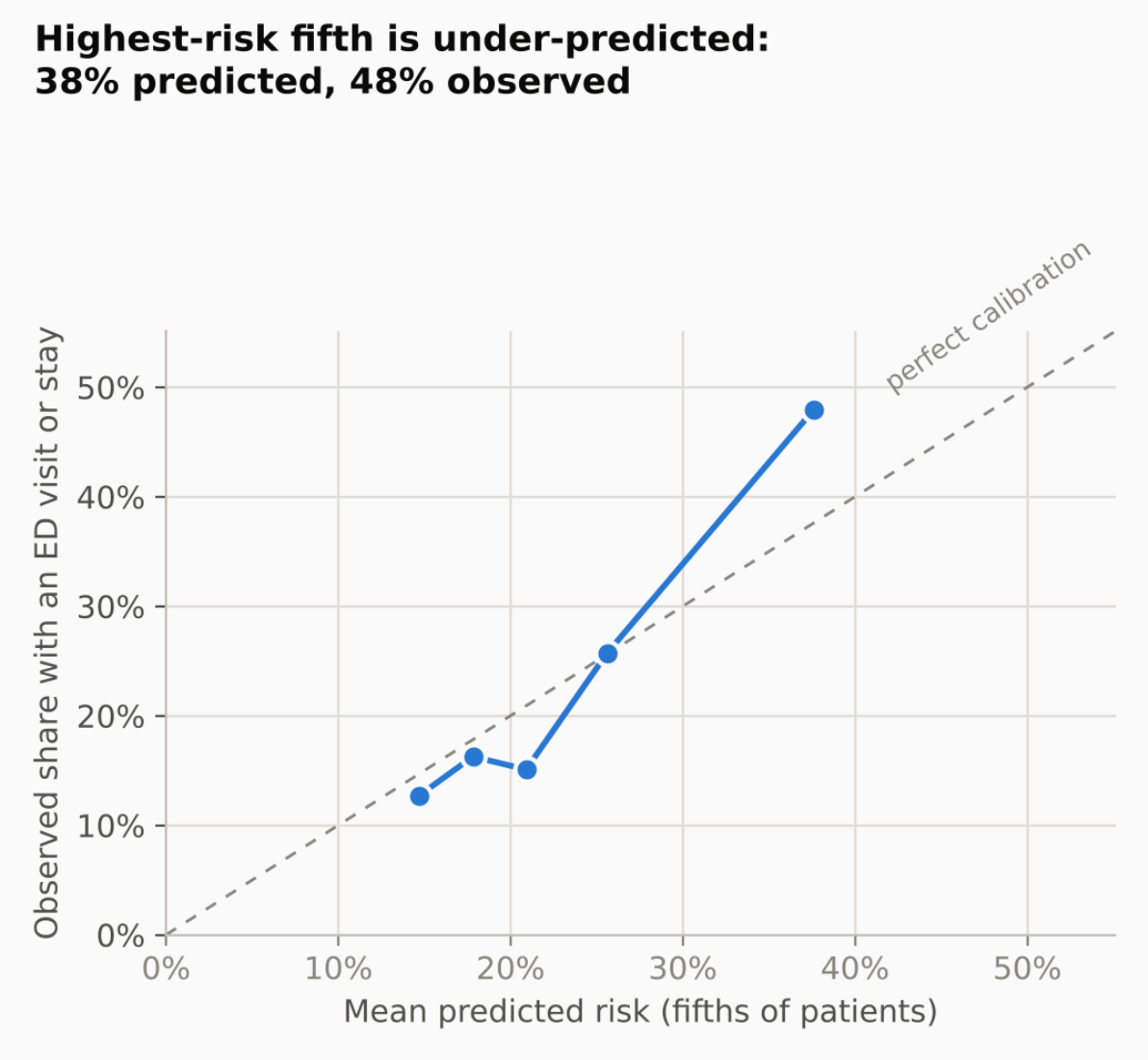
The prototype builds a diabetes registry from synthetic patients (about 1,700 registry patients across three simulated areas). It reports four care-gap measures, ranks patients on an outreach work list, and trains a model that predicts who will have an emergency visit or hospital stay in the next 12 months. Race and ethnicity are not model inputs. They are used only to audit the results.



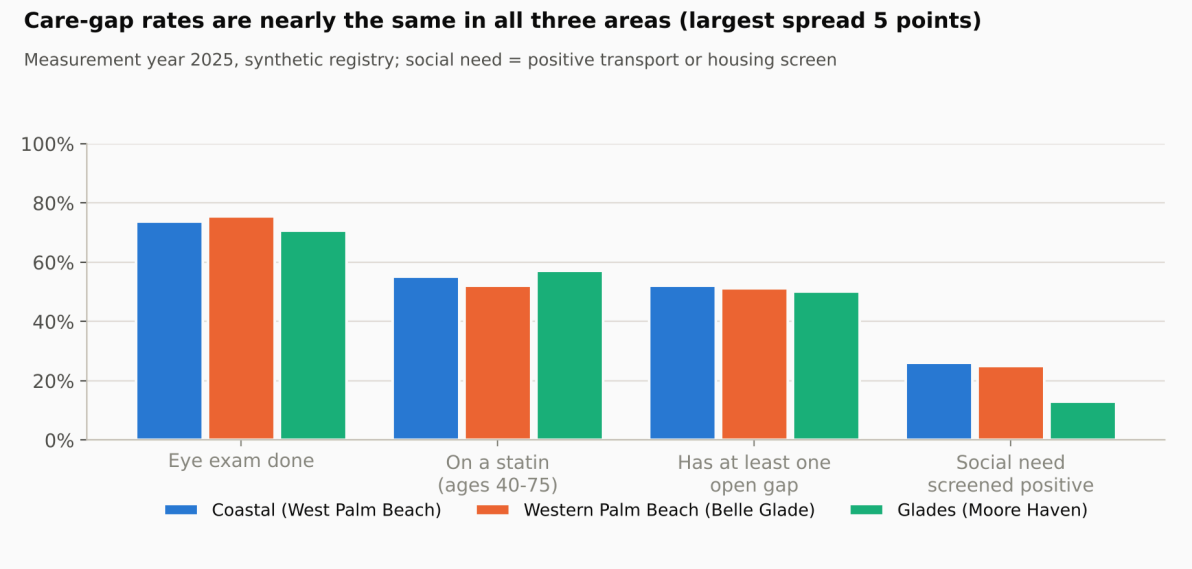
The best model reached AUC 0.69 against 0.64 for the best simple rule, a gain of 0.05 (paired 95% interval 0.02 to 0.08). A modest but real improvement.



Discrimination is similar across most groups and weakest for Asian and other patients (AUC 0.48, 97 patients), which is no better than chance.



Calibration is reasonable except in the highest-risk fifth, which is under-predicted (38% predicted, 48% observed).



Synthetic areas look alike on care gaps, so they cannot show rural access barriers.

What the equity audit found

- Flagging the top 20% of predicted risk caught 29% of later ED visits and hospital stays among privately insured patients, 45% among Medicare patients and 56% among uninsured patients.
- By area, the flag caught 35% in the coastal area, 50% in Belle Glade and 45% in Moore Haven.
- 51.7% of registry patients had at least one open care gap, and the work list had 890 patients (154 High, 216 Medium, 520 Low risk tier).

Retinopathy-screening scenario

Using `exam rate after = b + (1 - b) x r x c`, with reach r = 0.5 and completion c = 0.85, the eye exam rate rises 10 to 12 points from the synthetic baselines, or 23 points from an assumed 45% baseline. In Glades that means about 22 more patients screened from the synthetic baseline. These are illustrations of the formula, not predictions.

The registry prototype uses synthetic patients. It demonstrates method and does not describe any real patient or community.

Recommendations

> **Eastern and central Palm Beach.** For tracts ranked 2, 3, 4, 7 and 8 (Lake Worth Beach, West Palm Beach, Westgate), put outreach into coverage enrollment and sliding-fee scheduling. Uninsured adults run 24.8% to 41.5% there, yet each tract center is within 2 miles of a site, so access to a building is not the problem.

> **Western Palm Beach.** For tract 80.01 and its neighbors, bring screening to residents with pop-up or mobile clinics and rides. The area has 2 health center sites and no mobile units for about 33 thousand residents, and tract 80.01 is 11.3 miles from the nearest site.

> **Glades County.** Bring retinal screening to the clinic visit the patient already makes. In the scenario, reaching half of eligible patients with 85% completion lifts the eye exam rate by about 12 points from the synthetic baseline, or 23 points from an assumed 45% baseline.

Limits

- Patient-level results come from synthetic data and show method, not real clinical findings.
- PLACES values are model-based estimates and are uncertain in small tracts. The CDC does not publish transportation, food or housing insecurity for Florida tracts in this release, so SVI poverty and vehicle fields stand in.
- Distance is a straight line to HRSA sites. It omits other clinics, including the tribal health facility in Brighton.
- Glades has four tracts, so it is compared as an area and not ranked. One Palm Beach tract has no SVI rank and is not scored.
- The priority-score weights are a judgment call, tested with five alternative weightings.

Resume bullets

- Built a dashboard from public health data that shows where diabetes need is highest across 368 neighborhoods, and ranked the top 15 for outreach.

- Found that the barrier differs by place. In western Palm Beach it is cost (27% of adults uninsured). In Glades County it is distance (14.7 miles to the nearest health center site).
- Used AI (a machine learning risk model) to flag patients likely to need urgent care, so outreach staff can start with the highest-risk people. It outperformed a simple rule (AUC 0.69 vs. 0.64), tested on synthetic patient data.
- Checked the AI for fairness before use and found it caught more at-risk patients among uninsured patients (56%) than privately insured ones (29%), and failed for one racial group. Concluded that AI tools need an audit before they go live.
- Modeled how AI-assisted eye screening could close care gaps (about 12 percentage points in the scenario), and turned the results into a recommendation for each area.

Deliverables

DELIVERABLE	WHAT IT IS	WHERE
Project plan doc	Plan, key takeaways, executive summary, sources	Plan doc
Needs dashboard	Real-data tract map, three-area comparison, top-15 tracts, barrier profile, method and limits	Dashboard
Dashboard build script and data	Script that joins the public files and scores tracts, nine datasets, page source, rebuild README	`02_needs_dashboard/` in the zip
Registry prototype	Synthetic registry, risk model, equity audit, work list, screening scenario, four figures	`03_registry_prototype/` in the zip
Summary files	Key takeaways and executive summary	`01_plan_and_summary/` in the zip
Full package	Everything above in one download	`palm_beach_glades_portfolio.zip`

Sources

- CDC PLACES, 2025 release (tract and county; BRFSS 2023; model-based estimates)
- CDC/ATSDR Social Vulnerability Index, 2022 (Florida file; ACS 2018-2022)
- USDA Economic Research Service, RUCA codes 2020
- HRSA health center sites and primary care shortage designations (downloaded 2026-10-21)
- U.S. Census Bureau cartographic boundaries 2023 (tracts and places)